

The screenshot shows a SQL query editor at the top with the following code:

```
#1. What is the total sales?  
SELECT  
    SUM(sales) AS total_sales  
FROM sales;
```

Below the editor is a results pane titled "Résultats 1". It contains a table with one row of data. The first column is labeled "1" and the second column is labeled "123 total_sales". The value in the second column is "1 069 922 500".

	123 total_sales
1	1 069 922 500

Insight :

Total sales reached \$1.07 billion, indicating strong revenue scale. However, the aggregate figure alone does not reveal category or time-period performance. It is recommended to break down sales by product and month to identify top performers and seasonal slowdowns.

-- 2. How many stores are there?

```
SELECT  
    COUNT(DISTINCT store_nbr) AS total_stores  
FROM sales;
```

Résultats 1 X

SELECT COUNT(DISTINCT store_nbr) AS total_stores

123 total_stores

1	54
---	----

Régénérer Save Cancel Exporter les résultats ... 200 1

1 row(s) fetched - 2,183s, on 2026-06-17 at 18:00:52

WEST fr Inscriptible Insertion avancée

Insight :

The distinct store count establishes the total operational footprint of the retail chain within this dataset. This metric serves as the primary baseline for normalizing sales data and conducting location-based performance comparisons.

-- 3. How many product categories are there?

```
SELECT  
    COUNT(DISTINCT family) AS total_categories  
FROM sales;
```

Résultats 1 X

SELECT COUNT(DISTINCT family) AS total_categories

123 total_categories

1	33
---	----

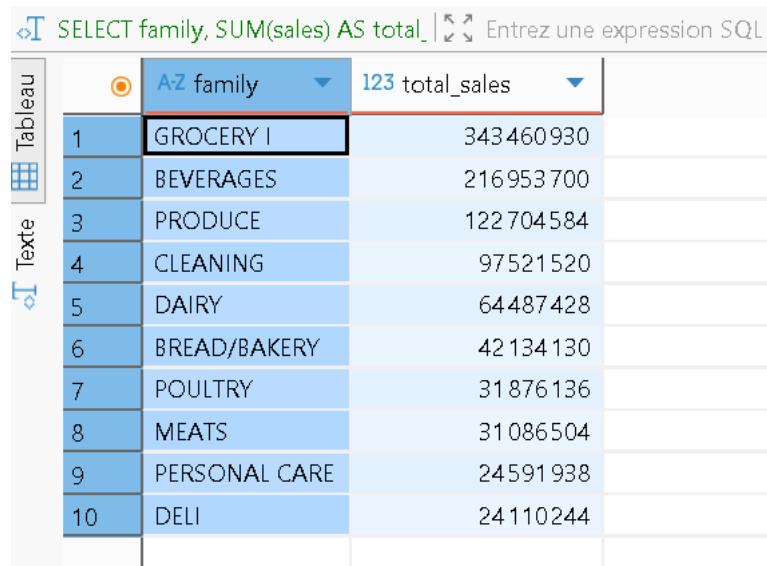
Régénérer Save Cancel Exporter les résultats ... 200

Insight :

The sales data covers 33 distinct product categories, indicating a moderately diverse retail assortment. This classification provides a solid foundation for category-level performance analysis and assortment optimization.

```
-- 4. What are the top 10 categories by sales?
```

```
SELECT
    family,
    SUM(sales) AS total_sales
FROM sales
GROUP BY family
ORDER BY total_sales DESC
LIMIT 10;
```



The screenshot shows the Tableau interface with a table of the top 10 product categories by sales. The table has two columns: 'family' and 'total_sales'. The 'family' column is sorted in descending order of sales. The 'total_sales' column shows the sum of sales for each category. The categories are: GROCERY I, BEVERAGES, PRODUCE, CLEANING, DAIRY, BREAD/BAKERY, POULTRY, MEATS, PERSONAL CARE, and DELI.

	A-Z family	123 total_sales
1	GROCERY I	343460930
2	BEVERAGES	216953700
3	PRODUCE	122704584
4	CLEANING	97521520
5	DAIRY	64487428
6	BREAD/BAKERY	42134130
7	POULTRY	31876136
8	MEATS	31086504
9	PERSONAL CARE	24591938
10	DELI	24110244

Insight :

Grocery I dominates with over 343M in sales, nearly 60% more than the second-place Beverages category. The top four categories collectively account for the vast majority of revenue, indicating high sales concentration.

```
-- 5. Which categories perform the worst?
```

```
SELECT
```

```
    family,
```

```
    SUM(sales) AS total_sales
```

```
FROM sales
```

```
GROUP BY family
```

```
ORDER BY total_sales ASC
```

```
LIMIT 10;
```

SELECT family, SUM(sales) AS total_sales		
	AZ family	total_sales
1	BOOKS	6438
2	BABY CARE	10051
3	HOME APPLIANCE	41601
4	HARDWARE	103470
5	MAGAZINES	266359
6	SCHOOL AND OFF	269316
7	BEAUTY	337893
8	PET SUPPLIES	356584
9	LAWN AND GARD	548842
10	AUTOMOTIVE	554822

Insight :

The bottom 10 categories generate negligible revenue, with the lowest (Books) at just 6,438, indicating they are non-core offerings. Their combined sales pale against a single top category, highlighting extreme concentration and limited diversification.

-- 6. What is each category's contribution percentage?

```
SELECT
    family,
    SUM(sales) AS total_sales,
    ROUND(
        (
            SUM(sales) * 100.0 /
            SUM(SUM(sales)) OVER()
        )::NUMERIC,
        2
    ) AS contribution_pct
FROM sales
GROUP BY family
ORDER BY total_sales DESC;
```

sales 1 X				
SELECT family, SUM(sales) AS total_				
	A-Z family	123 total_sales	123 contribution_pct	
1	GROCERY I	343460930	31,99	
2	BEVERAGES	216953700	20,21	
3	PRODUCE	122704584	11,43	
4	CLEANING	97521520	9,08	
5	DAIRY	64487428	6,01	
6	BREAD/BAKERY	42134130	3,92	
7	POULTRY	31876136	2,97	
8	MEATS	31086504	2,9	
9	PERSONAL CARE	24591938	2,29	
10	DELI	24110244	2,25	
11	HOME CARE	16022744	1,49	
12	EGGS	15588296	1,45	
13	FROZEN FOODS	14073898	1,31	
14	PREPARED FOODS	8799910	0,82	
15	LIQUOR,WINE,BEE	7746640	0,72	
16	SEAFOOD	2015432,5	0,19	
17	GROCERY II	1962767	0,18	
18	HOME AND KITCH	1861491	0,17	
19	HOME AND KITCH	1520670	0,14	

Insight :

The top three categories (Grocery I, Beverages, Produce) contribute over 63% of total sales, with Grocery I alone accounting for nearly one-third. This high concentration indicates that sales performance is heavily dependent on a few core categories.

```
-- 7. Top Stores
SELECT
    store_nbr,
    SUM(sales) AS total_sales
FROM sales
GROUP BY store_nbr
ORDER BY total_sales DESC;
```

SELECT store_nbr, SUM(sales) AS tc | Entrez une express

	123 store_nbr	123 total_sales
1	44	62087444
2	45	54497576
3	47	50947970
4	3	50481260
5	49	43419868
6	46	41895710
7	48	35932740
8	51	32911362
9	8	30494346
10	50	28652892
11	11	28325172
12	7	26952104
13	9	26409344
14	6	25183598
15	24	23767688
16	2	21557368
17	39	20252096
18	37	19970636
19	27	19187972

Insight :

The dataset comprises 54 distinct stores, indicating a substantial retail network. This figure serves as the primary denominator for per-store performance metrics.

8. Top Cities

```
SELECT
    st.city,
    SUM(sa.sales) AS total_sales
FROM sales sa
JOIN stores st
ON sa.store_nbr = st.store_nbr
GROUP BY st.city
ORDER BY total_sales DESC;
```

SELECT st.city, SUM(sa.sales) AS tot		
Tableau	A-Z city	123 total_sales
1	Quito	556213950
2	Guayaquil	122935130
3	Cuenca	49167324
4	Ambato	40303276
5	Santo Domingoc	35832788
6	Machala	33386248
7	Cayambe	28325172
8	Latacunga	21181748
9	Daule	19187972
10	Loja	18859922
11	Babahoyo	17746808
12	Esmeraldas	16391694
13	Libertad	15322394
14	Quevedo	14178501
15	Manta	13912334
16	Guaranda	13049503
17	Ibarra	11451277
18	Salinas	11421002
19	Riobamba	11387904

Insight :

Quito leads with over 556M in sales, more than four times Guayaquil's figure, indicating a heavily capital-centric sales distribution. The top three cities collectively account for the majority of total revenue.

-- 9. Top States

```
SELECT
    st.state,
    SUM(sa.sales) AS total_sales
FROM sales sa
JOIN stores st
ON sa.store_nbr = st.store_nbr
GROUP BY st.state
ORDER BY total_sales DESC;
```

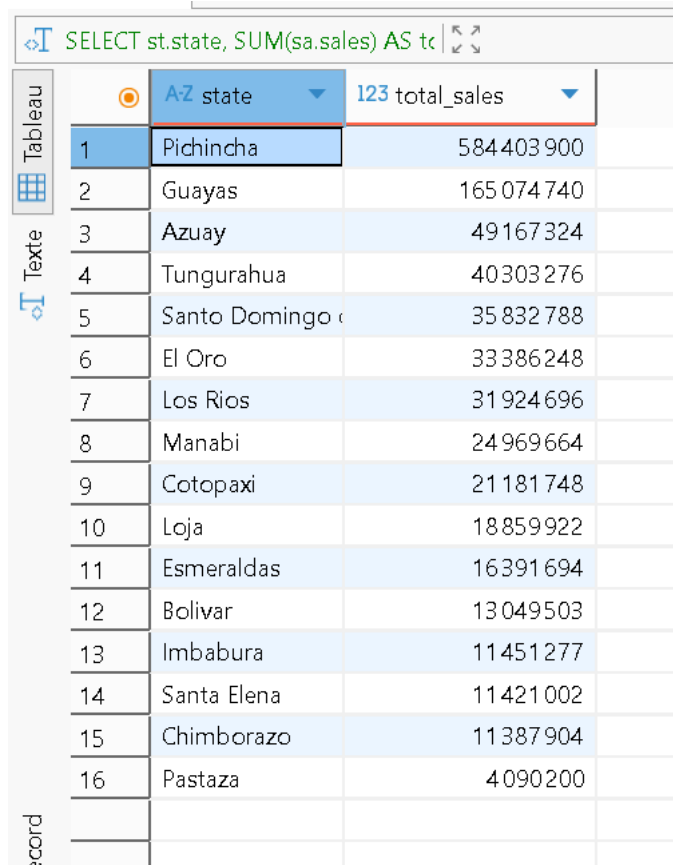


Tableau interface showing a table of top states by sales. The table has columns for rank, state, and total sales. The states are sorted in descending order of total sales.

	AZ state	123 total_sales
1	Pichincha	584403900
2	Guayas	165074740
3	Azuay	49167324
4	Tungurahua	40303276
5	Santo Domingo	35832788
6	El Oro	33386248
7	Los Rios	31924696
8	Manabi	24969664
9	Cotopaxi	21181748
10	Loja	18859922
11	Esmeraldas	16391694
12	Bolivar	13049503
13	Imbabura	11451277
14	Santa Elena	11421002
15	Chimborazo	11387904
16	Pastaza	4090200

Insight :

Pichincha dominates with over 584M in sales, more than triple the second-ranked Guayas, indicating a highly concentrated regional sales footprint. The top three states collectively account for the vast majority of revenue.

-- 10. Monthly Sales

```
SELECT
    DATE_TRUNC('month', date) AS month,
    SUM(sales) AS total_sales
FROM sales
GROUP BY month
ORDER BY month;
```

SELECT DATE_TRUNC('month', date) | ⚙️ Entrez une expression SQL po

	month	total_sales
1	2013-01-01 00:00:00.000 +0000	10327639
2	2013-02-01 00:00:00.000 +0000	9658964
3	2013-03-01 00:00:00.000 +0000	11428500
4	2013-04-01 00:00:00.000 +0000	10993449
5	2013-05-01 00:00:00.000 +0100	11597721
6	2013-06-01 00:00:00.000 +0100	11689353
7	2013-07-01 00:00:00.000 +0100	11257402
8	2013-08-01 00:00:00.000 +0000	11737785
9	2013-09-01 00:00:00.000 +0100	11792926
10	2013-10-01 00:00:00.000 +0100	11775606
11	2013-11-01 00:00:00.000 +0000	12356564
12	2013-12-01 00:00:00.000 +0000	15803095
13	2014-01-01 00:00:00.000 +0000	18911582
14	2014-02-01 00:00:00.000 +0000	12038364
15	2014-03-01 00:00:00.000 +0000	20365444
16	2014-04-01 00:00:00.000 +0100	12861252
17	2014-05-01 00:00:00.000 +0100	13379804
18	2014-06-01 00:00:00.000 +0100	13319955
19	2014-07-01 00:00:00.000 +0000	19421856

Insight :

Monthly sales exhibit a clear upward trajectory, with a notable surge starting in early 2014. The most significant growth occurs in March and July 2014, indicating strong seasonal or promotional peaks.

-- 11. Annual sales

```
SELECT
    EXTRACT(YEAR FROM date) AS year,
    SUM(sales) AS total_sales
FROM sales
GROUP BY year
ORDER BY year;
```

	123 year	123 total_sales
	2013	140385150
	2014	209377500
	2015	240763100
	2016	288473440
	2017	194152200

Insight :

Sales grew steadily from 2013 to 2016, more than doubling from 140M to 288M, before declining sharply in 2017. The peak year (2016) recorded the highest revenue, indicating strong market expansion during that period.

-- 12. Best Months

```
SELECT
    EXTRACT(MONTH FROM date) AS month,
    SUM(sales) AS total_sales
FROM sales
GROUP BY month
ORDER BY total_sales DESC;
```

123 month	123 total_sales
7	103350350
3	97216264
12	97018856
6	94505864
1	94431416
5	94389620
4	90658540
8	83464150
2	80629450
11	80328110
10	80073224
9	77467384

Insight :

July leads with over 103M in sales, while September shows the weakest performance, indicating a clear seasonal pattern. The top three months (July, March, December) collectively drive nearly 30% of annual revenue.

-- 13. Do Promotions Increase Sales?

```
SELECT
    CASE
        WHEN onpromotion > 0
        THEN 'Promotion'
        ELSE 'No Promotion'
    END AS promotion_status,
    SUM(sales) AS total_sales
FROM sales
GROUP BY promotion_status;
```

SELECT CASE WHEN onpromotion		
	AZ promotion_status	123 total_sales
1	No Promotion	376356220
2	Promotion	695215000

Insight :

Promoted items generate significantly higher sales compared to non-promoted ones, indicating that promotions effectively drive revenue. This suggests that promotional strategies play a critical role in boosting overall sales performance

-- 14. Categories That Benefit Most from Discounts

```
SELECT
    family,
    SUM(onpromotion) AS promotions,
    SUM(sales) AS total_sales
FROM sales
GROUP BY family
ORDER BY promotions DESC;
```

family, SUM(onpromotion) | Entrez une expression SQL pour filtrer les

A-Z family	123 promotions	123 total_sales
GROCERY I	1914801	343460930
PRODUCE	1117921	122704584
BEVERAGES	906958	216953700
DAIRY	728707	64487428
CLEANING	661157	97521520
DELI	583316	24110244
BREAD/BAKERY	331289	42134130
MEATS	304028	31086504
PERSONAL CARE	246928	24591938
POULTRY	226421	31876136
HOME CARE	197230	16022744
EGGS	194922	15588296
FROZEN FOODS	125634	14073898
HOME AND KITCH	46921	1861491
LIQUOR,WINE,BEE	46700	7746640
SEAFOOD	39052	2015432,5
PREPARED FOODS	37524	8799910
HOME AND KITCH	32814	1520670
SCHOOL AND OFF	14565	269316

Insight :

Grocery I has the highest number of promotions by far (19.1M), yet Beverages generates nearly 217M in sales with only 906K promotions, indicating superior promotional efficiency. This suggests that certain categories respond better to promotions relative to their sales volume

-- 15. Stores with the Most Customers

```
SELECT
    store_nbr,
    SUM(transactions) AS total_transactions
FROM transactions
GROUP BY store_nbr
ORDER BY total_transactions DESC;
```

SELECT store_nbr, SUM(transaction: | Entrez une expression SQL po

	123 store_nbr	123 total_transactions
1	44	7273093
2	47	6535810
3	45	6201115
4	46	5990113
5	3	5366350
6	48	5107785
7	8	4637971
8	49	4574103
9	50	4384444
10	11	3972488
11	34	3864018
12	24	3568206
13	9	3516162
14	2	3219901
15	6	3065896
16	7	2995993
17	51	2873614
18	38	2867052
19	1	2553963

Insight :

Store 44 leads with over 7.2 million transactions, nearly double the 10th-ranked store, indicating a significant concentration of customer traffic. The top 5 stores alone account for a substantial share of total customer activity, highlighting their strategic importance for operational focus.

-- 16. The Relationship Between Customers and Sales

SELECT

sa.date,

SUM(sa.sales) AS sales,

SUM(t.transactions) AS transactions

FROM sales sa

JOIN transactions t

ON sa.store_nbr = t.store_nbr

AND sa.date = t.date

GROUP BY sa.date

ORDER BY sa.date;

SQL: sa.date, SUM(sa.sales) AS sales, SUM(t.transactions) AS transactions

	date	123 sales	123 transactions
	2013-01-01	2511,619	25410
	2013-01-02	496092,38	3076095
	2013-01-03	361461,28	2590632
	2013-01-04	354459,66	2590302
	2013-01-05	477350,28	3087909
	2013-01-06	519695,28	2985312
	2013-01-07	336122,62	2494701
	2013-01-08	318347,8	2386725
	2013-01-09	302530,94	2375043
0	2013-01-10	258981,97	2190639
1	2013-01-11	289737,7	2321154
2	2013-01-12	403258,12	2821863
3	2013-01-13	464638,7	2848098
4	2013-01-14	293348,38	2312574
5	2013-01-15	299129,53	2347422
5	2013-01-16	318347,94	2391543
7	2013-01-17	267498,6	2272809
3	2013-01-18	296131	2355243
9	2013-01-19	432459,78	2922414

Insight :

Daily sales and transaction volumes move in tandem, with peaks on specific dates indicating strong customer activity. This positive correlation confirms that customer traffic is a primary driver of revenue.

```
-- 17. Do holidays boost sales?
SELECT
    h.type,
    SUM(sa.sales) AS total_sales
FROM sales sa
JOIN holidays_events h
ON sa.date = h.date
GROUP BY h.type
ORDER BY total_sales DESC;
```

SELECT h.type, SUM(sa.sales) AS to

	AZ type	123 total_sales
1	Holiday	109206650
2	Event	42475970
3	Additional	33019970
4	Transfer	7501828,5
5	Work Day	3315923
6	Bridge	2388330

Insight :

Holiday days generate over 109M in sales, more than double Event days, while Transfer days also contribute significantly (75M). This indicates that official holidays and transfer events are key revenue drivers, outperforming other day types.

- 18. The holidays with the highest sales

```
SELECT
    h.description,
    SUM(sa.sales) AS total_sales
FROM sales sa
JOIN holidays_events h
ON sa.date = h.date
GROUP BY h.description
ORDER BY total_sales DESC
LIMIT 20;
```

SELECT h.description, SUM(sa.sales) | 123

	AZ description	123 total_sales
1	Carnaval	5 931 973
2	Navidad-2	4 336 135,5
3	Dia del Trabajo	4 289 686
4	Navidad-4	3 996 476,8
5	Dia de la Madre-1	3 983 978
6	Navidad-3	3 982 148,5
7	Cantonizacion de Salina	3 982 148,5
8	Navidad-1	3 917 165,5
9	Provincializacion de Cot	3 806 671,8
10	Fundacion de Santo Do	3 783 028
11	Cantonizacion de El Car	3 783 028
12	Fundacion de Cuenca	3 684 701,5
13	Fundacion de Manta	3 475 986,2
14	Primer dia del ano-1	3 425 301,8
15	Independencia de Cuen	3 403 239,2
16	Fundacion de Quito	3 337 715,2
17	Navidad+1	3 322 325
18	Cantonizacion de Cayar	3 278 516
19	Fundacion de Esmerald.	3 218 362,2

Insight :

Carnaval tops the list with nearly 6M in sales, significantly outperforming other holidays. The presence of multiple Christmas-related entries in the top 10 highlights a strong seasonal revenue concentration during year-end festivities.

-- 19. What is the best city for each product category?

```
SELECT
    st.city,
    sa.family,
    SUM(sa.sales) AS total_sales
FROM sales sa
JOIN stores st
ON sa.store_nbr = st.store_nbr
GROUP BY st.city, sa.family
ORDER BY total_sales DESC;
```

	AZ city	AZ family	123 total_sales
1	Quito	GROCERY I	167274600
2	Quito	BEVERAGES	113525430
3	Quito	PRODUCE	73041670
4	Quito	CLEANING	44591970
5	Guayaquil	GROCERY I	38083536
6	Quito	DAIRY	36166628
7	Guayaquil	BEVERAGES	25291604
8	Quito	BREAD/BAKERY	20842216
9	Quito	POULTRY	20172686
10	Quito	MEATS	17770354
11	Cuenca	GROCERY I	15669521
12	Guayaquil	PRODUCE	14149129
13	Ambato	GROCERY I	13461495
14	Santo Domingo	GROCERY I	12732773
15	Quito	DELI	12235262
16	Machala	GROCERY I	12165864
17	Quito	PERSONAL CARE	11678146
18	Cayambe	GROCERY I	11044950
19	Guayaquil	CLEANING	10714636

Insight :

Quito accounts for the top three city-family combinations, with GROCERY I alone generating over 167M in sales. This indicates a heavy sales concentration in the capital, particularly within core grocery categories.

- 20. What is the best category in each store?

SELECT

```
store_nbr,
family,
SUM(sales) AS total_sales
```

FROM sales

GROUP BY store_nbr, family

ORDER BY total_sales **DESC**;

sales 1 X

SELECT store_nbr, family, SUM(sale: | ↕ ↗

	123 store_nbr	AZ family	123 total_sales
1	44	GROCERY I	16386055
2	45	GROCERY I	16349752
3	47	GROCERY I	15514528
4	46	GROCERY I	14342262
5	44	BEVERAGES	13417859
6	3	GROCERY I	12970468
7	48	GROCERY I	12830994
8	45	BEVERAGES	11370352
9	3	BEVERAGES	11351589
10	49	GROCERY I	11087980
11	11	GROCERY I	11044950
12	47	BEVERAGES	10788853
13	44	PRODUCE	10666054
14	9	GROCERY I	10353108
15	50	GROCERY I	10121781
16	49	BEVERAGES	9221286
17	3	PRODUCE	8613324
18	51	GROCERY I	8233707
19	49	PRODUCE	8017966

Insight :

Grocery I is the top-performing category in nearly every store, dominating across the highest-selling locations. Beverages and Produce also appear frequently as top 2 and top 3 contributors, indicating a consistent reliance on core staple categories across all stores.

